Benha University
Faculty of Engineering (Shoubra)
Electrical Engineering

ELE123 Technical Reports

ELE114 Technical Report Writing

(2026-2027)

Dr. Christina Gamal

Basic Report Structure

Lecture 2

A report usually has the following components:

- Cover Page
- Title page
- Abstract
- Dedication*
- Disclaimer*
- Acknowledgement*
- Table of contents
- List of Tables
- List of Figures
- Nomenclature*
- Introduction chapter
- Central Chapters
- Conclusions
- References
- Appendices*
- Biography*

Preliminary Pages

Main Text Pages

Supplement Pages

* Optional Pages

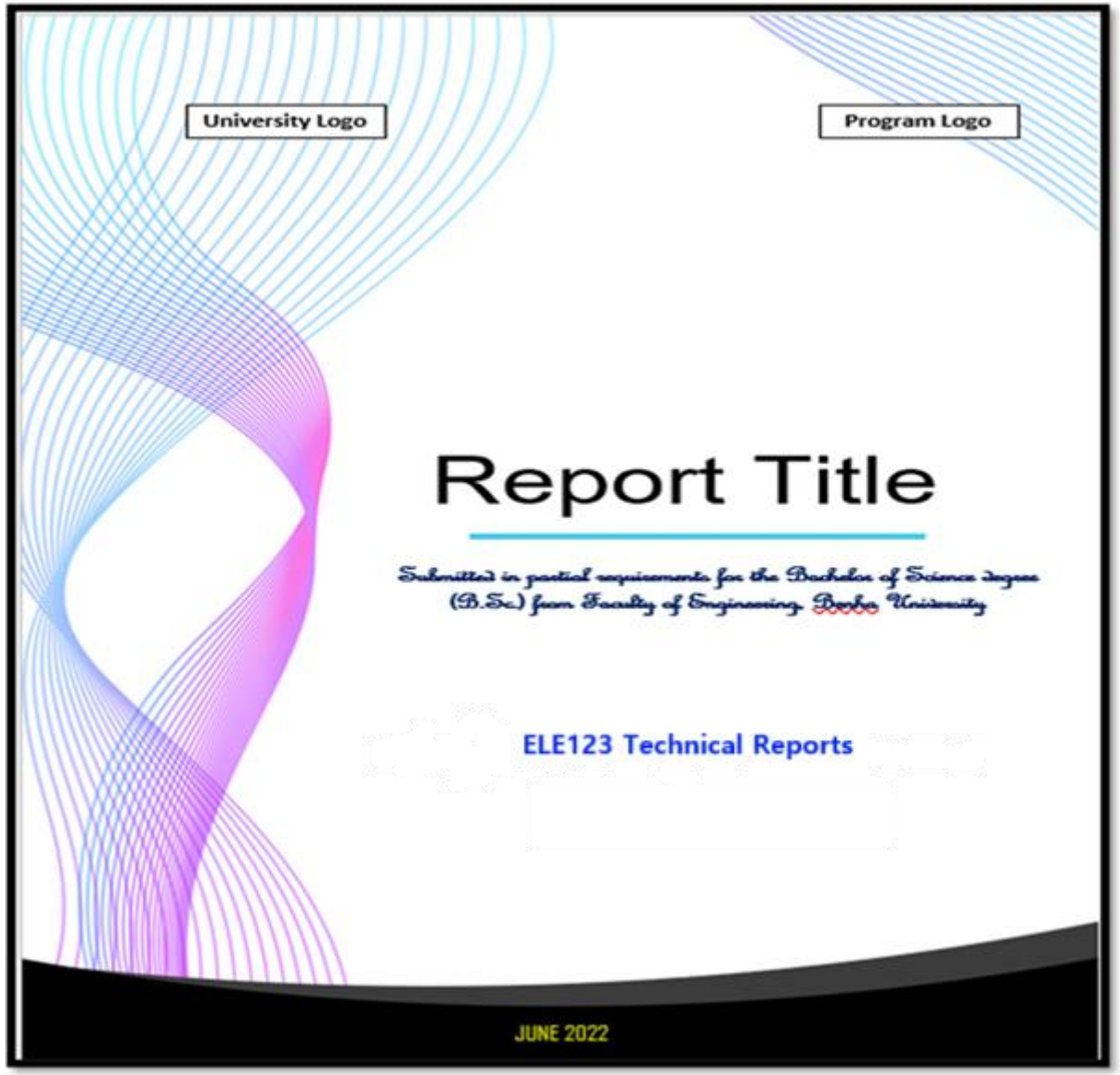
1- Cover Page

The purpose of the cover page is to identify the whole report. The cover page should be **simple** and **attractive**. It should contain the following information:

- University name & Faculty name with logo
- Subject name and code
- Project (report) title
- Date of submission

➢ **Cover page** can include author's name (and affiliation).

➢ **Cover page** can be used in place of or in addition to a title page depending on the document's purpose and its type.



- The **title of the report** must be considered carefully. A good title is striking and clearly reflects the contents of the report. The title of the report should indicate exactly what the report is about. The reader should know not only the general topic, but also the specific aspect of the topic addressed in the report.

Compare the following pairs of report titles:

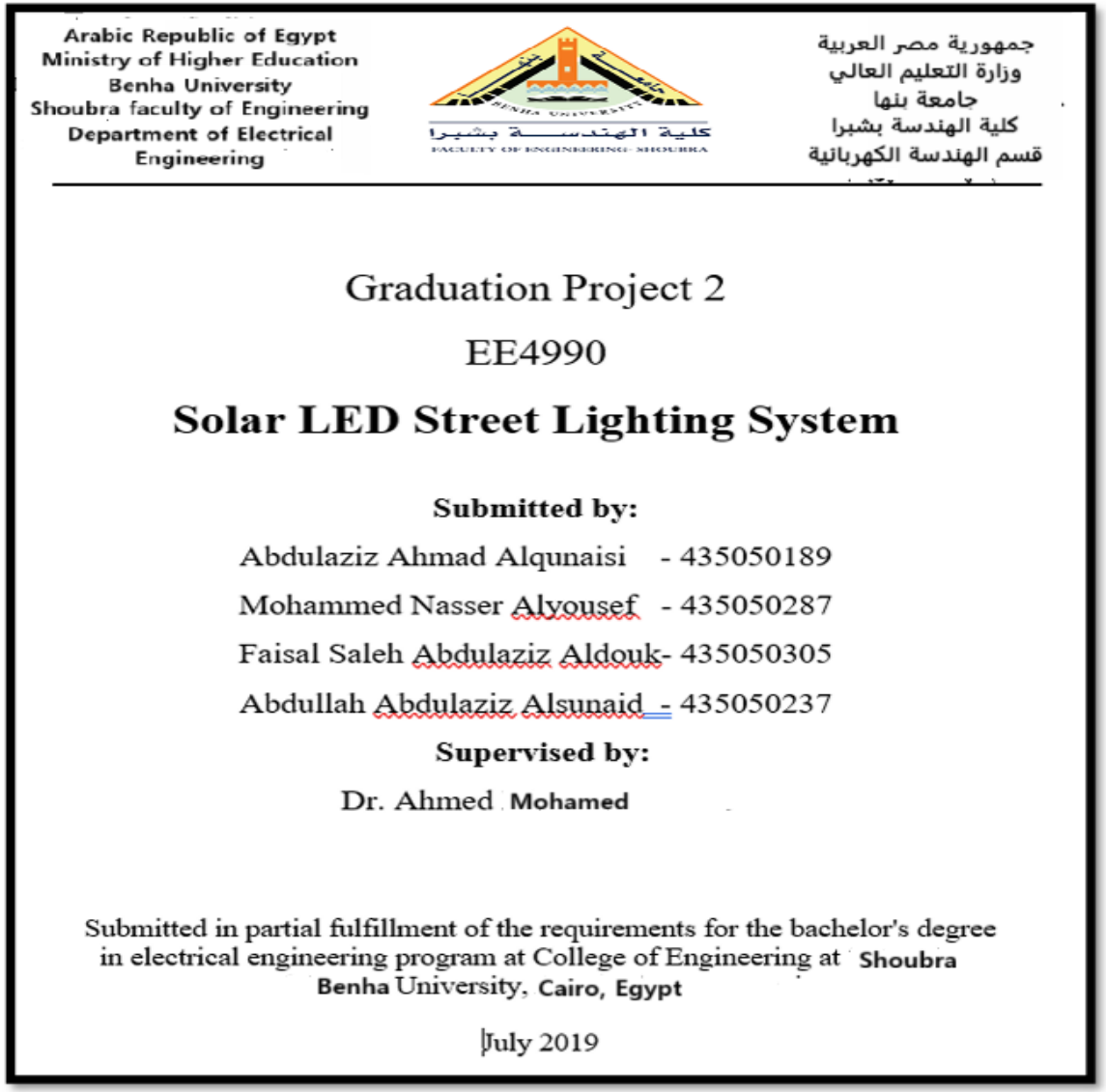
Weak titles	Strong titles
Transformer Analysis	Analysis of a Single-Phase Transformer Under Different Loads
Internet-Based ATIS	An evaluation of Internet based Automated Traveller Information Systems

Weak titles	Strong titles
Assignment 1	Assignment 1: Water consumption data collection
Lab Report 1	Laboratory Report 1: Experimental Determination of Transformer Parameters

2- Title Page

The title page contains all information given on the cover page with white background with extra information:

- Student names and ID
- Supervisor name



Arabic Republic of Egypt
Ministry of Higher Education
Benha University
Shoubra faculty of Engineering
Department of Electrical Engineering

جمهورية مصر العربية
وزارة التعليم العالي
جامعة بنها
كلية الهندسة بشبرا
قسم الهندسة الكهربائية

Graduation Project 2

EE4990

Solar LED Street Lighting System

Submitted by:

Abdulaziz Ahmad Alqunaisi - 435050189

Mohammed Nasser Alyousef - 435050287

Faisal Saleh Abdulaziz Aldouk- 435050305

Abdullah Abdulaziz Alsunaid - 435050237

Supervised by:

Dr. Ahmed Mohamed

Submitted in partial fulfillment of the requirements for the bachelor's degree in electrical engineering program at College of Engineering at Shoubra Benha University, Cairo, Egypt

July 2019

3- Abstract

- Abstracts highlight major points of the report and explain why your work is important.
- Abstract is not an introduction to the report. It often provides no background information.
- Abstract is also known as an overview, or short summary.
- The abstract is often written last as its purpose is to provide a summary of the report's essential information.
- All material in the abstract will also be explained in more details through the report.
- The abstract should appear on a separate page after the title page, and it is usually about 100–200 words in length.

The abstract should include the following elements:

- why? purpose of the report (Purpose) P
- how? brief details of the approach/procedure/methods (Method) M
- what? important results/findings (Results) R
- so what? major conclusion(s) and recommendation(s) (Conclusion) C

When you are writing your abstract, you should avoid:

- Referring to other works

  (e.g., “According to Smith, 2020…”).

- Defining any terms

  (e.g., “Bandwidth is defined as…”).

- Adding information that isn't contained in the larger work

4- Dedication

- This is a short sentence, in the middle of a separate page, in which the report is dedicated to a family member, or a friend. It may be left out and is seldom included in short technical reports. It is more suited to theses.

5- Disclaimer

A ‘disclaimer’ or declaration of authenticity is often required in major pieces of work such as large reports, projects, and theses. It is a **signed statement** declaring that the report is the work of the stated author(s). The disclaimer should appear on a page following the abstract.

This is an example of a disclaimer statement:

I declare the following to be my own work, unless otherwise referenced, as defined by the University’s policy on plagiarism … followed by the author’s signature.

6- Acknowledgments

The acknowledgments page is optional; however, it is essential that significant assistance, editing, or work carried out by another person or organization be acknowledged. Also, students may wish to thank colleagues or supervisors. Also, we must acknowledge institutions that provided money or made facilities available.

This is an example of an acknowledgment statement:

I would like to thank my supervisor, Dr. ....., Benha University for his encouragement and guidance throughout the project. Also, I would like to express deep thanks to Dr. ....., Cairo University for his help in setting up my experiments.

7- Table of Contents

- The table of contents must begin on a new page.
- The page is provided with a heading, such as “Contents” or “Table of Contents”, followed by a list of the three main levels of headings and their page numbers.
- Journal papers do not have a table of contents.
- The first item in the table of contents should be the first heading that appears after the table of contents, for example List of Figures.
- Front matter that precedes the table of contents are not listed.
- Appendices must be listed, each with their title and starting page.

TABLE OF CONTENTS

Acknowledgements..... vi

List of Figures.....viii

List of Tables..... x

Chapters

1. INTRODUCTION..... 1

1.1 Basics Of Solar Cells.....3

1.2 Physics Of Solar Cells.....7

1.3 Solar Array's.....10

2. CHARACTERISTICS OF A PV MODULE..... 12

2.1 Types Of Solar Panels..... 12

2.2 Electrical Connections Of The Modules.....15

8- List of Tables and List of Figures

- These lists, arranged according to the table and figure number,
- Each begins on a new page and indicates the relevant page number in the right-hand column.
- The titles of tables and figures must be descriptive enough so that a specific figure or table can be identified in the list and must correspond to the title used for the figure or table in the text.

List of Figures

Figure No.	Title	Page
1.1	Applications of solar LED lighting system	3
1.2	Block Diagram Solar LED Street Lighting System	4
1.3	Solar LED Street Light structure	5
2.1	Photoelectric effect	8
2.2	Silicon crystal lattice	8
2.3	Solar radiation paths.	9
2.4	Definition of AM.	10
2.5	Illustration of AM1.5 path for a surface at 37◦ tilt.	10

9- Nomenclature

- The list of symbols that are used must begin on a new page. The list is arranged in the following sequence:
  - All the **ordinary symbols** are listed first, followed by the **superscripts** and then the **subscripts**. Finally, the **auxiliary symbols**, for example overbar and underscore for vectors and averages or accent marks for time-dependent components, are listed.
- The following order must be used within each of these groups:
  - First, all the **Roman letters** (in alphabetical order, with the capital letter of each symbol before the small letter, for example “A” followed by “a”, followed by “B”); Then all the **Greek symbols** (in the order of the Greek alphabet, capital letters before small letters);

10- Introduction

- The introductory chapter should provide the reader with the following information:
  - The context in which the report originated, how it links to or differs from preceding or related work, and the limitations that were placed on the work.
  - The purpose of the report.
  - The motivation for the work or report, that is, why the work was undertaken.
  - The organization of the report.
- The introduction also contains a general overview of previous work in the field and definitions of words or expressions that have a specific meaning in the document. This is what is called a **literature review**. Literature is used to list the textbooks and journal articles related to the report topic; it can also include official publications such as engineering standards and government reports.

11- Central Chapters

- The structure of the central chapters depends on the contents of the report.
- The following are general **guidelines** for the central chapters:
  - Every chapter should be focused on one topic; i.e., it should have a clear purpose. The chapter title normally reflects the purpose.
  - The contents of the central chapters must remain strictly linked to the purpose of the report. Contents that are only marginally important should preferably be placed in the appendices.

- The following **structure of chapters** or within chapters can usually be followed and corresponds to a scientific approach:

  - Introduction: the purpose of the chapter and how it links to the purpose of the report;
  - Underlying or simplified assumptions;
  - Analytical or numerical theory used, or the procedure for the investigation;
  - Measured results, results of the analysis or observations;
  - Processing of results;
  - Interpretation of results;
  - Conclusions: usefulness and importance of results; how the results contribute to achieving the purpose of the report.

12- Conclusions

- The conclusion of a report must be related to, and resulting from, the material which appears in the report.
- it must not introduce any new material.
- Quite often, the **present tense** is used.
- Conclusions are quite often read by managers before the main text of the report and hence, should summarize the main points clearly. This section also may include:
  - Aim(s) and objective(s) of the report,
  - application(s) of results,
  - limitations and advantages of the findings,
  - judgement/evaluation of the author(s).
- The conclusion does NOT:
  - provide background information on the topic
  - explain the motivation for the project
  - refer to figures, tables, or references contained in the report.

Conclusion

1- What is the Problem?

2-How did they solve it?

3- What is new about their method?

4- How did they test the method ?

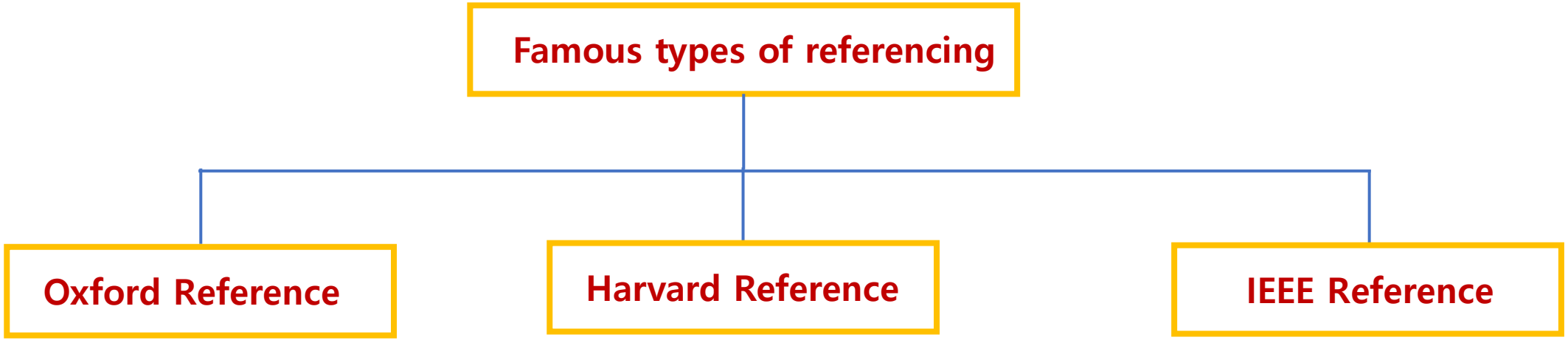
5- What were the results?

A simple but effective approach is presented for extracting unknown parameters of PV modules using the single-diode model based on three remarkable points provided in the manufacturer's datasheet. This is achieved by formulating a closed-form expression as a function of module series resistance and solving numerically while increasing the value of the ideality factor (a) in small increments. The effectiveness of the formulation presented in the study is demonstrated by comparing the results alongside those from reference characteristics provided by manufacturers of the BP SX-150 and MSX-60 PV modules. The results show a close correlation in terms of the unknown parameters. Moreover, compared with similar computations in the literature, the computation results from our study for both PV modules produced better outcomes.

6- Why is the method important?

13- Referencing

- In-text citations are used throughout the report to acknowledge the sources of information. The full references for the citations are then listed at the end of the report in the Reference list.



Institute of Electrical and Electronics Engineers

Oxford Reference:

[1] Taha Selim Ustun, Cagil Ozansoy, and Aladin Zayegh, "Modeling of a Centralized Microgrid Protection System and Distributed Energy Resources According to IEC 61850-7-420", IEEE Transactions on Power Systems, 2012, 27 (3), 1560-1565.

In text citation

superscript numbers in the text to indicate references

e.g., ^1

as discussed previously[1]

In Reference list (footnotes (bottom of the page) or **endnotes** (at the end of the document))

Author(s), title (in quotation marks), journal/conference, year, volume(number), page numbers.

Harvard Refefence:

[2] Taha Selim Ustun, Cagil Ozansoy, and Aladin Zayegh, (2012) Modeling of a Centralized Microgrid Protection System and Distributed Energy Resources According to IEC 61850-7-420, IEEE Transactions on Power Systems, 27 (3), 1560-1565.

In text citation

(last author name, year of publication)

e.g., (Ustun et al., 2012)

In Reference list

Author(s), year (in parentheses), title (no quotation marks), journal name, volume (number), page numbers.

IEEE Referencing:

[3] T. S. Ustun, C. Ozansoy, and A. Zayegh, "Modeling of a centralized microgrid protection system and distributed energy resources according to IEC 61850-7-420," *IEEE Transactions on Power Systems*, vol. 27, no. 3, pp. 1560–1565, 2012.

In text citation

[Number] in the text to indicate reference's number

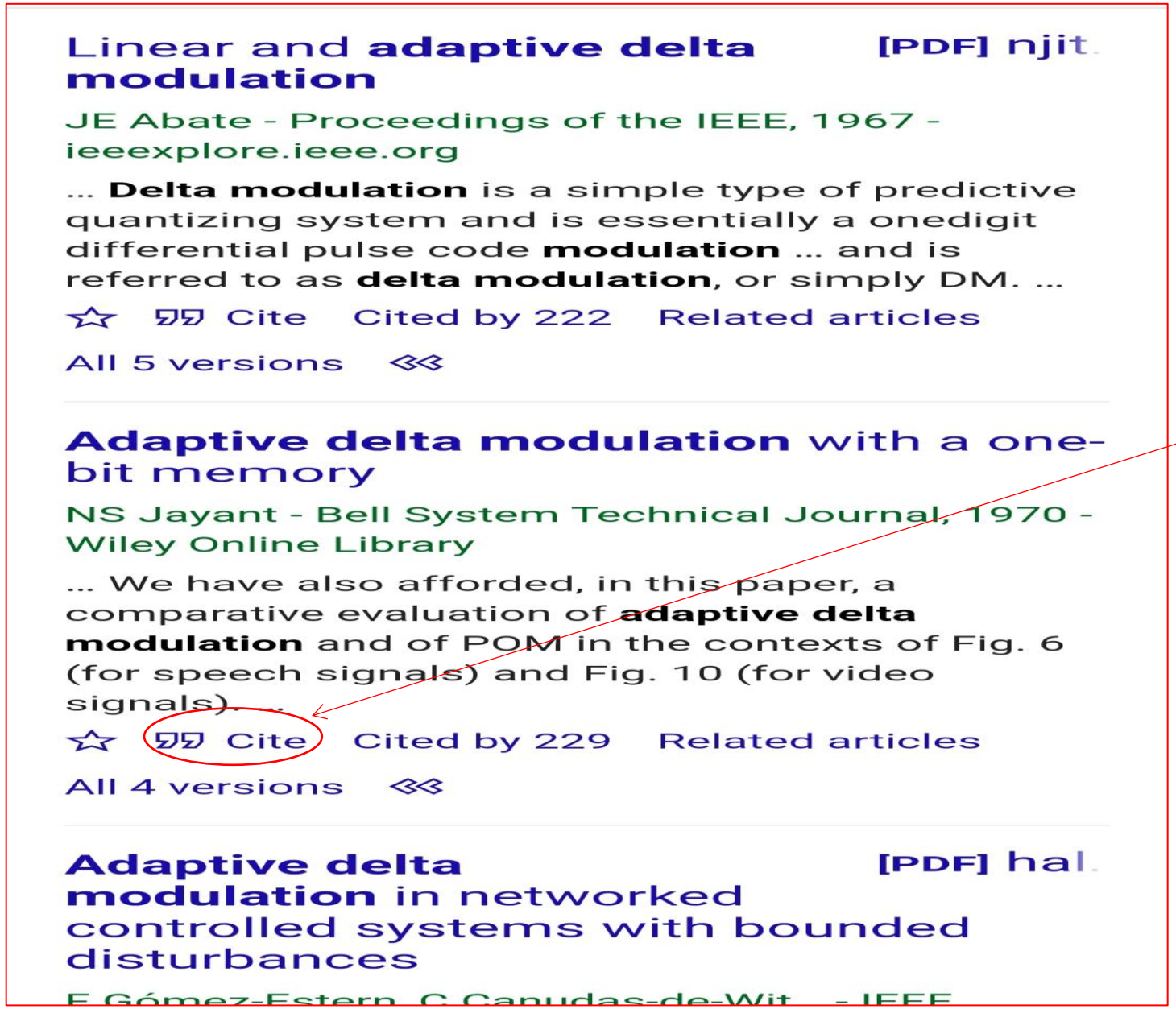
e.g., [3]

as discussed previously [3]

In Reference list

Author(s), title (in quotation marks), journal/conference, volume, number, page numbers, year.

How to cite a reference



Click here

Cite

MLA

Abate, John Edward. "Linear and adaptive delta modulation." *Proceedings of the IEEE* 55.3 (1967): 298-308.

APA

Abate, J. E. (1967). Linear and adaptive delta modulation. *Proceedings of the IEEE*, *55*(3), 298-308.

Chicago

Abate, John Edward. "Linear and adaptive delta modulation." *Proceedings of the IEEE* 55, no. 3 (1967): 298-308.

Harvard

Abate, J.E., 1967. Linear and adaptive delta modulation. *Proceedings of the IEEE*, *55*(3), pp.298-308.

Vancouver

Abate JE. Linear and adaptive delta modulation. Proceedings of the IEEE. 1967 Mar;55(3):298-308.

BibTeX EndNote RefMan RefWorks

14- Appendix

- Explanatory material that would interfere with the logical continuity.
- Supporting material not critical and may be of interest only to a few readers.
- Supplementary calculations; detailed derivations; datasheets; charts, maps, graphs as additional information; detailed experimental results; description of equipment; description of software tool; Each item must have an identifying tag and a title.

• Each appendix must be labelled with a number (or letter) and title
• The appendix numbers and titles must be listed on the Contents page under the heading Appendices (if more than one) or Appendix (if only one)
• Each appendix must be referred to by number (or letter) at the relevant point in the text.

15- Biography

- short description of the author(s) of the report.
- It tells the reader *who wrote the report* and gives some background on their education, research interests, or professional experience.

End of Lecture